



Advanced NLP

Session 1 - Introduction, History of NLP & Paradigm Shifts in LLMs

Teacher: Dr. Behrouz Minaei-Bidgoli

Presenter: Dr. Behrouz Minaei-Bidgoli

Presentation creator: Poorya Saneei

Contents

1. Course Overview
2. History of Natural Language Processing



1.

Course Overview

Faculty & Lead



Prof. B. Minaei-Bidgoli

Course Instructor

T

Dr. Najmeh Torabian

Course Manager

Course Website

Check the official course website for lectures, assignments, and resources:

just-nlp.ir

To join the Telegram group, message
@PooryaSN on Telegram.

Teaching Assistants & Mentors



Poorya Saneei (Head TA)

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Course Schedule & Lecture Roadmap

PART 1: FOUNDATIONS & ARCHITECTURES (W1-W4)

S1 Introduction, History of NLP & Paradigm Shifts in LLMs

[Lecture](#) • W1 (06-28) • Dr. Minaei

S2 Word Representations, Distributed Semantics & Embeddings

[Lecture](#) • W1 (06-30) • Fateme Marefat

S3 Neural Network Basics & Backpropagation in NLP

[Lecture](#) • W2 (07-04) • Alireza Nazari

S4 Autoregressive Language Modeling & Pre-training Objectives

[Lecture](#) • W2 (07-06) • Alireza Nazari

S5 Recurrent Sequence Architectures: RNNs, LSTMs & GRUs

[Lecture](#) • W3 (07-11) • Alireza Nazari

S6 Attention Mechanisms & The Transformer Architecture

[Lecture](#) • W3 (07-13) • Setare Kahnamuee

Q1 In-Class Quiz 1 (Covering Sessions S1 to S6)

Quiz 1 • W4 (07-18) • TAs

PART 2: ADVANCED MODELING & OPTIMIZATION (W4-W7)

S7 Subword Tokenization (BPE, SentencePiece) & Multilingual

[Lecture](#) • W4 (07-20) • Fateme Marefat

S8 LLM Pre-training: Data Pipelines, Filtering & Curation

[Lecture](#) • W5 (07-25) • Setare Kahnamuee

S9 Scaling Laws & In-Context Learning (ICL)

[Lecture](#) • W5 (07-27) • Setare Kahnamuee

S10 Parameter-Efficient Fine-Tuning (PEFT: LoRA, QLoRA) & Distillation

[Lecture](#) • W6 (08-02) • Saba Sattari

S11 Decoding Algorithms, Sampling Strategies & Chain of Thought

[Lecture](#) • W6 (08-04) • Mohammad Khaki

S12 Advanced Reasoning & Mathematical Planning in LLMs

[Lecture](#) • W7 (08-09) • Mohammad Khaki

Q2 In-Class Quiz 2 (Covering Sessions S7 to S12)

Quiz 2 • W7 (08-11) • TAs

Course Schedule & Lecture Roadmap (Part 2)

PART 3: ADVANCED MULTIMODALITY & GENERATION (W8-W11)

S13 Retrieval Augmented Generation (RAG) & Dense Retrieval (DPR)

[Lecture](#) • W8 (08-16) • Mohadeseh Emadi

S14 Vision-Language Models (VLM: CLIP, LLaVA) & Multimodality

[Lecture](#) • W8 (08-18) • Poorya Saneei

S15 Evaluation Techniques, Benchmarks & LLM Leaderboards

[Lecture](#) • W9 (08-23) • Mohadeseh Emadi

S16 Graph Neural Networks for Natural Language Processing (GNN for NLP)

[Lecture](#) • W9 (08-25) • Poorya Saneei/ Setare Kahnemuee

S17 Diffusion & Flow Matching Models in NLP Generation Tasks

[Lecture](#) • W10 (08-30) • Poorya Saneei

S18 Fundamentals of Reinforcement Learning in NLP

[Lecture](#) • W10 (09-02) • Saba Sattari

Q3 In-Class Quiz 3 (Covering Sessions S13 to S17)

[Quiz 3](#) • W11 (09-07) • TAs

PART 4: RL, ALIGNMENT & EFFICIENT INFERENCE (W11-W14)

S19 Post-Training Alignment: RLHF & Direct Preference Optimization (DPO)

[Lecture](#) • W11 (09-09) • Poorya Saneei

S20 LLM-Based Agents, Function Calling & Tool Use

[Lecture](#) • W12 (09-14) • Mohammad Khaki

S21 Model Quantization (AWQ, GPTQ) & Efficient Inference serving

[Lecture](#) • W12 (09-16) • Mohammad Khaki

S22 Mixture of Experts (MoE) & Scaling Sequence Length (FlashAttention)

[Lecture](#) • W13 (09-21) • Mohadeseh Emadi

S23 Process Supervision & Test-Time Compute Scaling

[Lecture](#) • W13 (09-23) • Saba Sattari

S24 Interpretability, Safety, Bias & Adversarial Attacks in LLMs

[Lecture](#) • W14 (09-28) • Fateme Marefat

S25 Research Skills, Experimental Design & Open Horizons in NLP

[Lecture](#) • W14 (09-30) • Setare Kahnemuee

EXTRA1 & EXTRA2

Reserve & Buffer Sessions - 2 Days

[W15 \(10-05\)](#) AND [W15 \(10-07\)](#) • Contingency & Catch-up

FINAL

Final Exam • Wednesday, 1405-10-30 • 11:00-13:00

• Covers Sessions S13-S25

Grading Policy & Breakdown

PROJECTS & QUIZZES

Multi-Phase Course Project

Hands-on phased capstone project released in stages across the semester

45%

Quizzes (Q1, Q2, Q3)

Three progress review quizzes (totaling 20%): Q1 (Week 4, covers S1–S6), Q2 (Week 7, covers S7–S12), and Q3 (Week 11, covers S13–S18)

20%

EXAMS & EVALUATIONS

Final Exam

Comprehensive examination covering theoretical and applied topics from Sessions S13 to S25 (Wednesday, 1405-10-30 • 11:00–13:00)

30%

In-Class Participation

Active discussion, paper reviews, and classroom engagement

5%

Total Available Score

100%

2.

History of Natural Language Processing

Main Resource For this Section: <https://web.stanford.edu/class/cs224n/>

What is Natural Language Processing (NLP)?

Core Definition

An advanced field of technology that enables computers to process, generate, and interact with human language (such as text or speech) to bridge the gap between human communication and machine understanding.

01

Learn Useful Representations

Capture complex linguistic meaning in a structured, mathematical format. These dense representations (such as word embeddings) can be effectively utilized for key downstream tasks, including document classification and semantic analysis.

02

Generate Expressive Language

Produce coherent, contextually relevant language—spanning text, dialogue, and code. This capability powers advanced user interactions including real-time translation, dynamic conversational agents, and automated question answering.

03

Bridge Language and Action

Translate text and instructions into functional execution. This enables systems to execute complex tasks, solve logical problems, and directly interact with virtual environments, such as code generation and manipulation inside an IDE.

Four Eras of NLP & Entwined Fields

Era 01

1940–1969**Early Explorations**

Pioneering research, early machine translation attempts, and foundational formal language theory.

Era 02

1970–1992**Symbolic NLP**

Hand-built symbolic NLP systems featuring increasing linguistic and mathematical formalization.

Era 03

1993–2012**Statistical NLP**

Empirical shift toward statistical or probabilistic NLP, leading into general supervised machine learning.

Era 04

2013–Now**Deep Learning**

Neural networks, unsupervised or self-supervised large language models, and reinforcement learning.

Only Somewhat Entwined Fields**NLP / Computational Linguistics**

Focused on human language computation, formal syntax, and linguistic semantics.

AI / Neural Networks

Focused on general learning algorithms, optimization, and cognitive architectures.

Machine Translation & The Origins of NLP

HISTORICAL SEED

Translation as a Problem in Cryptography

“Also knowing nothing official about, but having guessed and inferred considerable about, the powerful new mechanized methods in cryptography—methods which I believe succeed even when one does not know what language has been coded—one naturally wonders if the problem of translation could conceivably be treated as a problem in cryptography. When I look at an article in Russian, I say: ‘This is really written in English, but it has been coded in some strange symbols. I will now proceed to decode.’”



Warren Weaver

American Mathematician &
Communication Pioneer

*(1954:18, quoting a letter he wrote in
1947)*

Machine Translation: The Origin of NLP

WEAVER'S SEED

Warren Weaver (March 1947)

Translation as Cryptography

"When I look at an article in Russian, I say: 'This is really written in English, but it has been coded in some strange symbols. I will now proceed to decode.'" – Warren Weaver, March 1947



Warren Weaver

Mathematician & pioneer science funder at the Rockefeller Foundation. Coauthored the foundational Information Theory work with Claude Shannon.

THE SKEPTICISM

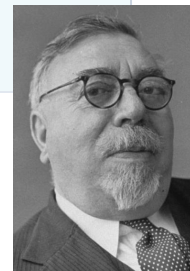
Norbert Wiener (April 1947)

The Limits of Quasi-Mechanical Schemes

"...as to the problem of mechanical translation, I frankly am afraid that the [semantic] boundaries of words in different languages are too vague ... to make any quasi-mechanical translation scheme very hopeful."

Norbert Wiener

MIT originator of cybernetics, seeking to tie together communication, control, and feedback systems in both living organisms and machines.



Machine Translation: The Origin of NLP

1954 DEMONSTRATION

Georgetown-IBM Public Debut

By ROBERT A. PLUMB

A public demonstration of what is believed to be the first successful use of a machine to translate meaningful texts from one language to another took place here yesterday afternoon.

This may be the culmination of centuries of search by scholars for "a mechanical translator." So far the system has a vocabulary of only 250 words. But there are no foreseeable limits to the number of words that the device can store or the number of languages it can be directed to translate.

Scholars and scientists who worked on it believe that within a few years the system may greatly increase communication, particularly in technical subjects, by making translation quick, accurate and easy.

scientists of the corporation and scholars of the Georgetown University Institute of Languages and Linguistics in Washington.

The "mechanical" part of the translation system, which is mostly electronic, is a standard commercial model of the largest International Business Machines "stock" computer. This device, called the IBM Type 701 Electronic Data Processing Machine, was put on the market last April. Since then twelve of the machines have been sold to commercial, military and university computation laboratories.

The "literary" part of the system is a mechanical model of language devised at Georgetown by Prof. Leon Dostert and Dr. Paul Garvin. The corporation's share in the project was conducted by

THE TECHNOLOGY

IBM 701 Electronic Calculator

Calculator Takes on a New Job: Language Translation



An electronic calculator produced by the International Business Machines Corporation was demonstrated yesterday in a new role: translating Russian phrases into English. The device has a "vocabulary" of 250 Russian words and can be adapted to other languages. Mrs. Marilyn Poffe "types" Russian phrases on a I. B. M. punch cards that are fed into the machine.

This card is punched with a sample Russian language sentence (in interpreting at the top) in standard BSI punched-card code. It is then accepted by the TOL converter into its own binary language through the means of stored elementary and operational syntactical programs into the English language equivalent which is then printed.

THE QUALITY OF COAL IS DETERMINED BY

Also, students work and read below a table with instructions. I used colored circles

The early history of NLP: MT in the 1950s

GEORGETOWN-IBM DEMO

1954 Public Debut

First public demonstration of machine translation

“The Georgetown-IBM demonstration was the first public demonstration of machine translation... translating Russian sentences into English using a card-programmed computer.”

Key Milestones

- Used a carefully selected vocabulary of 250 words
- Translated basic chemistry and political texts
- Sparked massive funding and enthusiasm for MT

THE ELECTRONIC BRAIN

Russian to English Translation

Paramount News Highlights



AI Begins: The Dartmouth Summer Research Project 1956



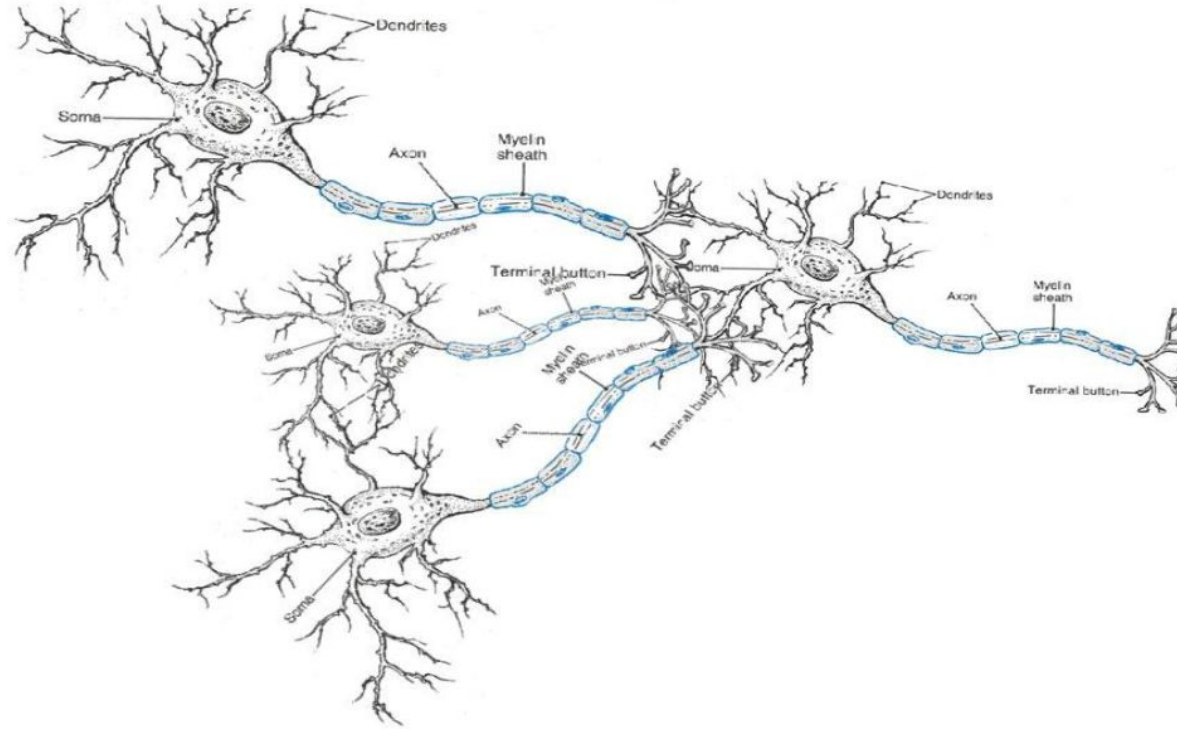
HISTORICAL MILESTONE

The Birth of Artificial Intelligence

The Dartmouth Summer Research Project is widely considered the founding event of AI as an academic discipline.

- **Key Organizers:** John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon.
- **Coining the Term:** The proposal officially introduced and defined the term "Artificial Intelligence."
- **The Core Conjecture:** That every aspect of learning or intelligence can in principle be so precisely described that a machine can be made to simulate it.

Neural Networks: The inspiration from knowledge of neurons



Foundations of Neural Networks: McCulloch & Pitts

Warren S. McCulloch

1898–1969

Neurophysiologist (MD), psychiatrist, and cyberneticist.

Moved to **MIT in 1951**, establishing a hub for brain science.

His productive collaboration with Pitts ended abruptly after the **Wiener split in 1952**.

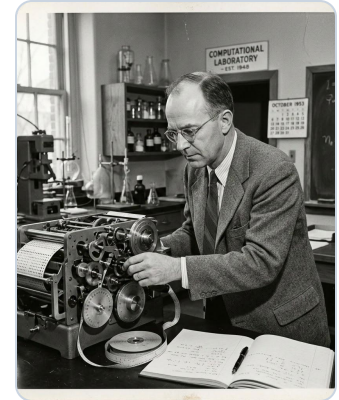
Walter Pitts

1923–1969

Self-taught logician and computational neuroscientist.

Escaped a tough background to live and study at **Chicago and MIT**.

A brilliant scholar of immense intellect, his life sadly ended early due to **alcoholism**.



The Breakthrough: A Logical Calculus of Ideas Immanent in Nervous Activity (1943)

This seminal paper introduced the **first mathematical model of a neural network**. McCulloch and Pitts mathematically proved that a network of simplified binary threshold neurons could perform any logical operation and **implement a Turing Machine**, laying the dual foundations for both computational neuroscience and modern artificial intelligence.

McCulloch-Pitts Neuron

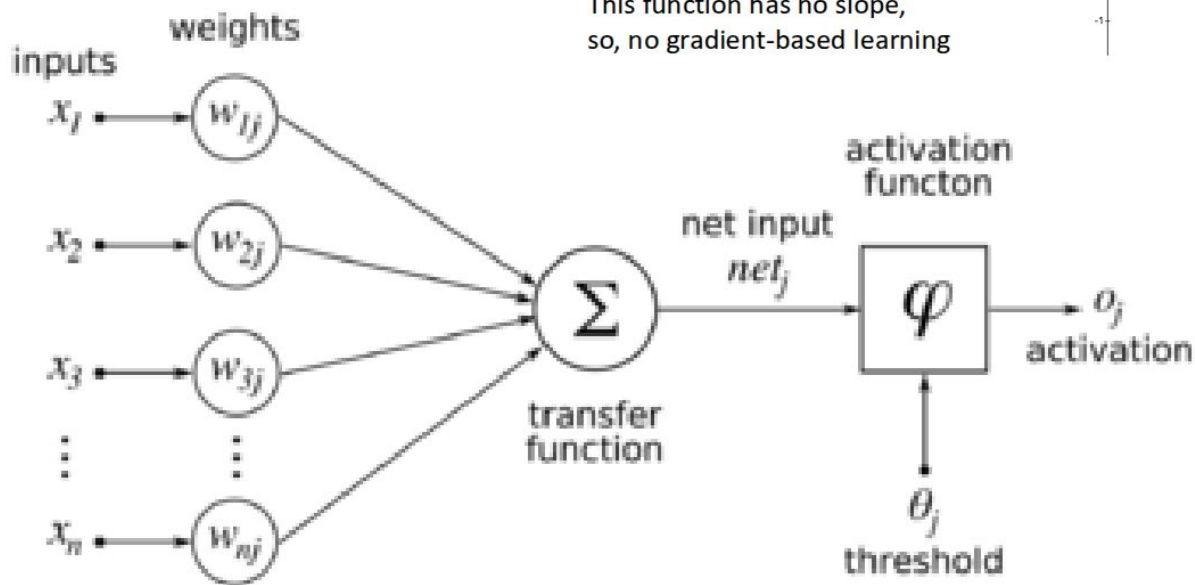
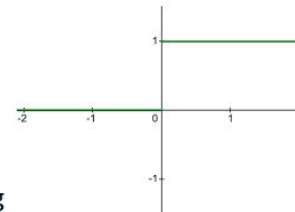
McCulloch-Pitts Neuron

Original McCulloch & Pitts

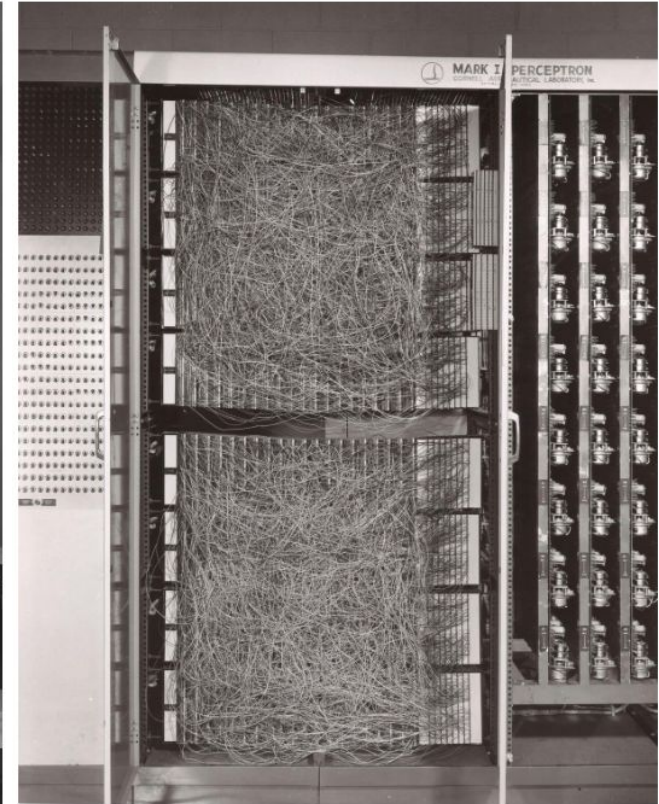
1943 threshold unit:

$$\begin{aligned} & \mathbf{1}(Wx > \theta) \\ &= \mathbf{1}(Wx - \theta > 0) \end{aligned}$$

This function has no slope,
so, no gradient-based learning



Frank Rosenblatt: The (Mark I) Perceptron



Early AI Hype! New York Times July 8, 1958

New York Times

TUESDAY, JULY 8, 1958

NEW NAVY DEVICE LEARNS BY DOING

Psychologist Shows Embryo of Computer Designed to Read and Grow Wiser

The Navy revealed the embryo of an electronic computer today that it expects **will be able to walk, talk, see, write, reproduce itself and be conscious of its existence.**

The embryo—the Weather Bureau's \$2,000,000 "704" computer—learned to differentiate between right and left after fifty attempts in the Navy's demonstration for newsmen.

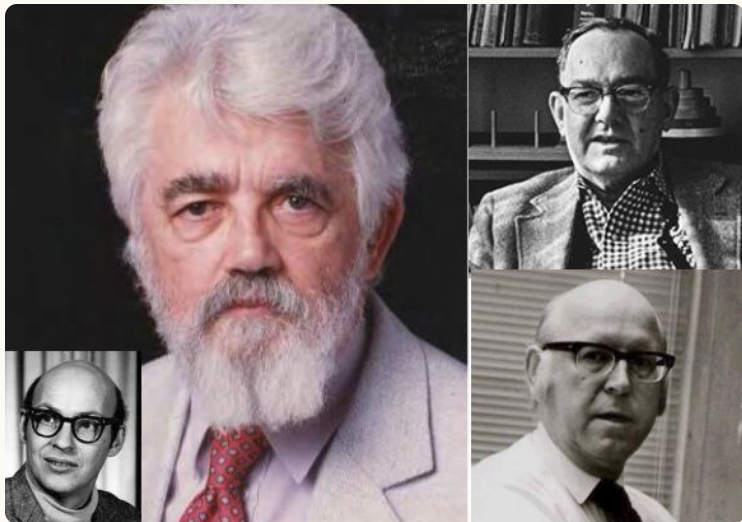
< Key Takeaways & Historical Context

- **The Perceptron's Debut:** Frank Rosenblatt's neural network model was demonstrated running on an IBM 704 vacuum-tube mainframe computer.
- **Sensationalist Claims:** The Navy funded press release boldly predicted the machine would eventually acquire human-like motor skills and full self-consciousness.
- **The First "AI Spring":** This extreme optimism defined the earliest hype cycle, showcasing that dramatic expectations have run parallel with AI development since its inception.

Two visions of artificial intelligence in the 1950s

(Symbolic) Artificial Intelligence

Minsky, McCarthy, Simon, Newell



Cybernetics

Wiener, Rosenblatt





Information Retrieval: Vannevar Bush

The Atlantic

JULY 1945

AS WE MAY THINK

*“Consider a future device for individual use, which is a sort of mechanized private file and library... **memex** will do. A memex is a device in which an individual stores all his books, records, and communications, and which is mechanized so that it may be consulted with exceeding speed and flexibility.”*

The Memex: An enlarged intimate supplement to memory.

Vannevar Bush: Key Credentials

Dean of Engineering at MIT

Established a foundation of rigorous technical leadership and educational excellence.

President of Carnegie Institution

Led major scientific research initiatives and directed institutional investments in pioneering projects.

WWII Wartime Leadership

Chairman of the **National Defense Research Committee**.

Directed immense funding to guide wartime science, including coordination for the Manhattan Project.

Information Retrieval: Cyril Cleverdon

The Cranfield Tests

1957 – 1967

THE DAWN OF BENCHMARKING

“Defined the fundamental methodology of benchmark tests for language, establishing a standardized framework that remains the cornerstone of modern system evaluation.”

Methodological Core:

- Structured Document Collection
- Standardized User Queries
- Definitive Correct Answers (Ground Truth)

Cyril Cleverdon: Key Credentials

Librarian & IR Pioneer

Librarian at Cranfield College of Aeronautics; systematically revolutionized how search systems are measured.



Exhaustive Data Collection

Meticulously compiled exhaustive answers on document relevance over a tightly controlled, small corpus.

Legacy and Impact

The Cranfield paradigm established that retrieval systems could be evaluated quantitatively, paving the direct path for modern NLP and search engine benchmarks.

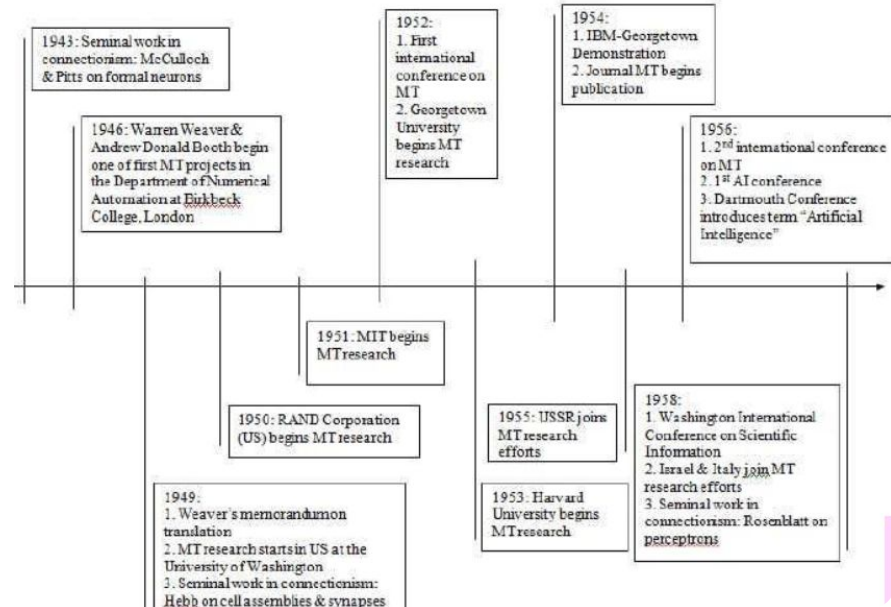
The 1950s origins of NLP/Computational Linguistics

Timeline Overview

This detailed historical map traces the critical academic, institutional, and technical breakthroughs that shaped the field of NLP and Computational Linguistics during its foundational decade.

Source Citation

Ruth Camburn, "A Short History of Computational Linguistics" (CSU Fresno, 2013).



1960s: David Glenn Hays: ACL founder and dependency parsing

Key Contributions

ACL Founder

- Led the founding of the Association for Computational Linguistics (ACL) in the 1960s.
- Served as a central figure in establishing the field's professional community.

Dependency Parsing Pioneer

- Developed early, foundational mathematical models of dependency grammars.
- Authored influential early works on computer-based parsing algorithms.



Raj Reddy (1937–): AI & NLP Pioneer

Biographical Milestones

Stanford University

- PhD Student (1963–1966)
- Assistant Professor (1966–1969)

Carnegie Mellon University

- AI & Robotics Pioneer & Faculty Member

Turing Award Laureate

- Honored for groundbreaking work in AI systems



2. Hand-built demonstration NLP systems

Increasing Formalization

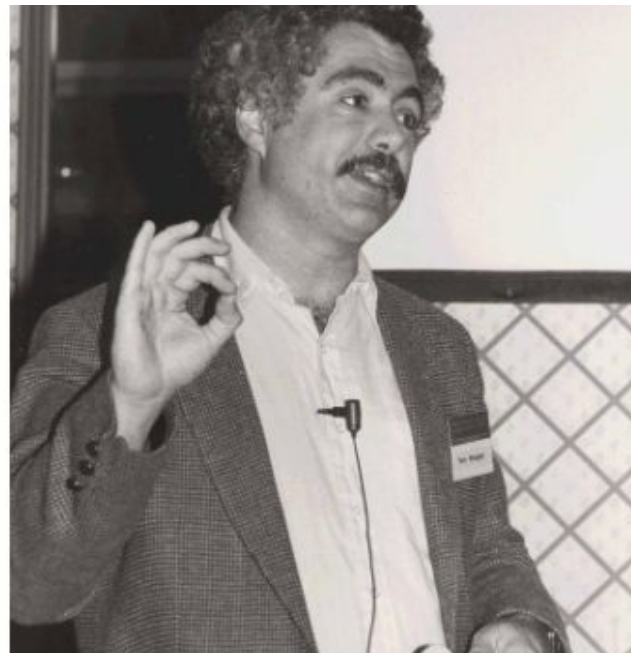
1970–1992

Terry Winograd

Academic Profile

1946— · Stanford University

- **Stanford Faculty:** Served as a distinguished member of the Stanford faculty from 1973 to 2014.
- **AI / NLP Period:** Contributed pioneering research in Artificial Intelligence and Natural Language Processing from 1973 to 1984.



Formal and Unification-based Grammars for NLP: Martin Kay

Martin Kay

1935–2021 · Stanford University

- **Functional Unification Grammar (FUG):** Introduced unification formalisms to represent syntactic structures.
- **Chart Parsing:** Invented efficient tabular parsing algorithms, crucial for handling ambiguous natural language.
- **Computational Pioneer:** Shaped the foundations of machine translation and modern computational linguistics.



Constructing the meaning of human language: 1967–2017

01. Tokenization

Tokenize a sentence into words:

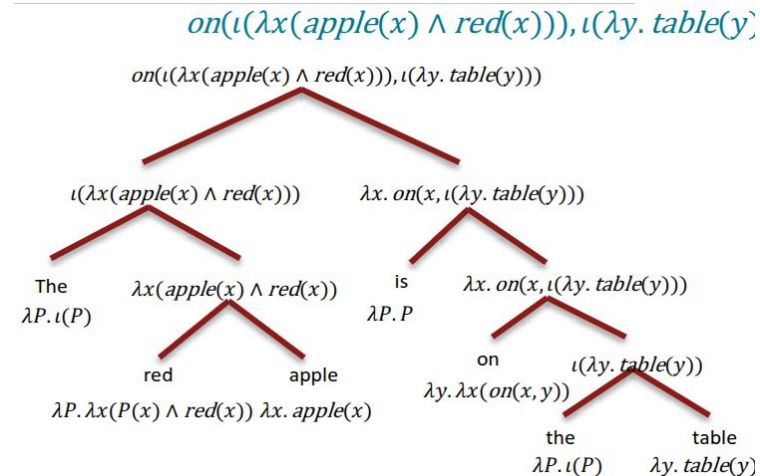
"The red apple is on the table"

02. Syntactic Parsing

Parse the tokenized words into a hierarchical tree or graph data structure using context-free grammars and beyond.

03. Semantic Composition

Construct its meaning via **lexical lookup**, then **semantic composition**, using a "rule-to-rule" approach working up the tree (e.g., PP: $\alpha(\beta) \rightarrow P: \alpha$ NP: β).



Semantic Parse Tree Visualization

Mapping sentences to lambda calculus representation

NLP and Knowledge Representation: Norvig (1986) Ph.D.

Peter Norvig

1986 Ph.D. Thesis · UC Berkeley

"A Unified Theory of Inference for Text Understanding"

The Language Analyzed:

"...a young boy named Chang Lee lived with his widowed mother. Every day, little Chang bravely set off with his net, hoping to catch a few fish from the sea, which they could sell..."

Key Inferences Resolved:

- **Semantic Context:** The net is a fishing net; the sea surrounds the island.
- **Pronoun Resolution:** "they" refers to Chang and his mother.
- **Anaphora:** "which" refers to the fish.



The unified theory of inference

“As we have just seen, a suitable knowledge base is a prerequisite for making proper inferences” (p. 4). It’s built to enable inferences.

System had 6 general forms of inference (2 pairs, so 4 basic types):

01. Elaboration

Filling a slot to connect two entities.

e.g., John got piggy bank [REASON] have money [REASON] buy present

02. Reference

Hey – it’s coreference!!!

Resolving multiple expressions that refer to the same entity.

03. View Application

Viewing a situation through a metaphorical lens.

e.g., "The Red Sox killed the Yankees" (KILLED is viewed as DEFEAT-CONVINCINGLY)

04. Concretization

Inferring a specific action from a general concept.

e.g., TRAVELLING in an AUTOMOBILE is an instance of DRIVING

NLP in the 1970s and 1980s

Speech Recognition & Stochastic Modeling

Foundational work on speech recognition using hidden Markov models and the "noisy channel" (ideas that later revolutionized NLP).

Rules-Driven AI & Parsing

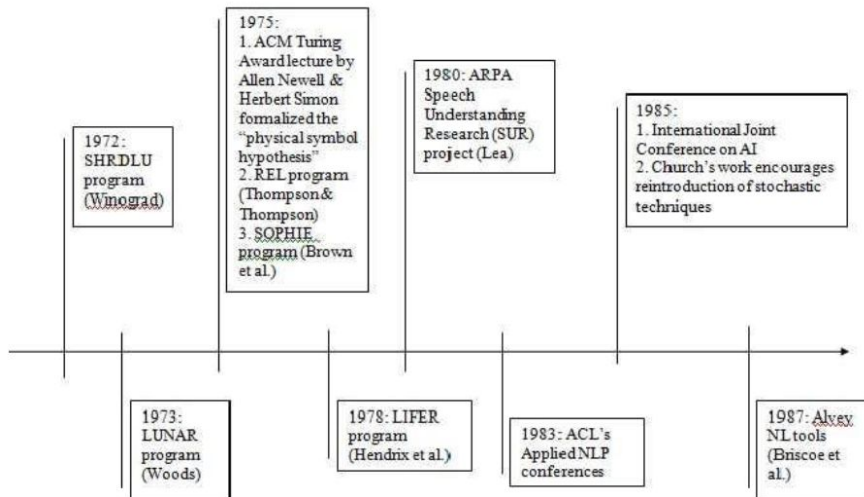
Logic programming and deterministic algorithms for syntactic parsing (e.g., LFG).

Natural Language Understanding

Growing experimental interest in early NLU systems like SHRDLU, LUNAR, and CHAT-80.

The "AI Winter"

Symbolic AI approach hits the wall, entering a period of reduced funding and interest.

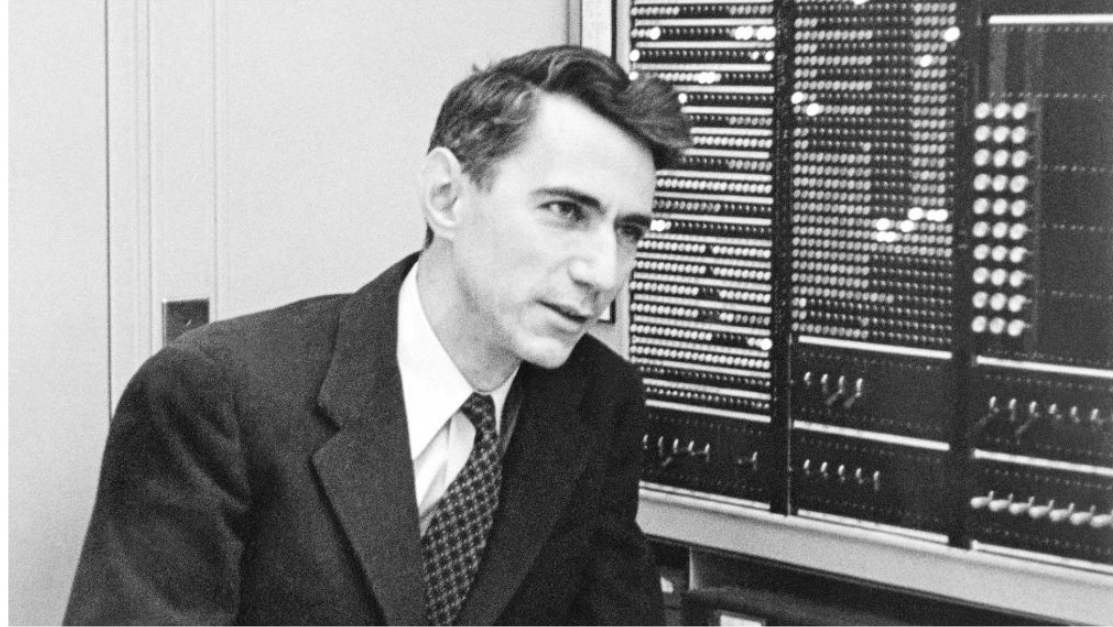


3. Statistical or Probabilistic NLP (“StatNLP”)

Supervised ML for NLP

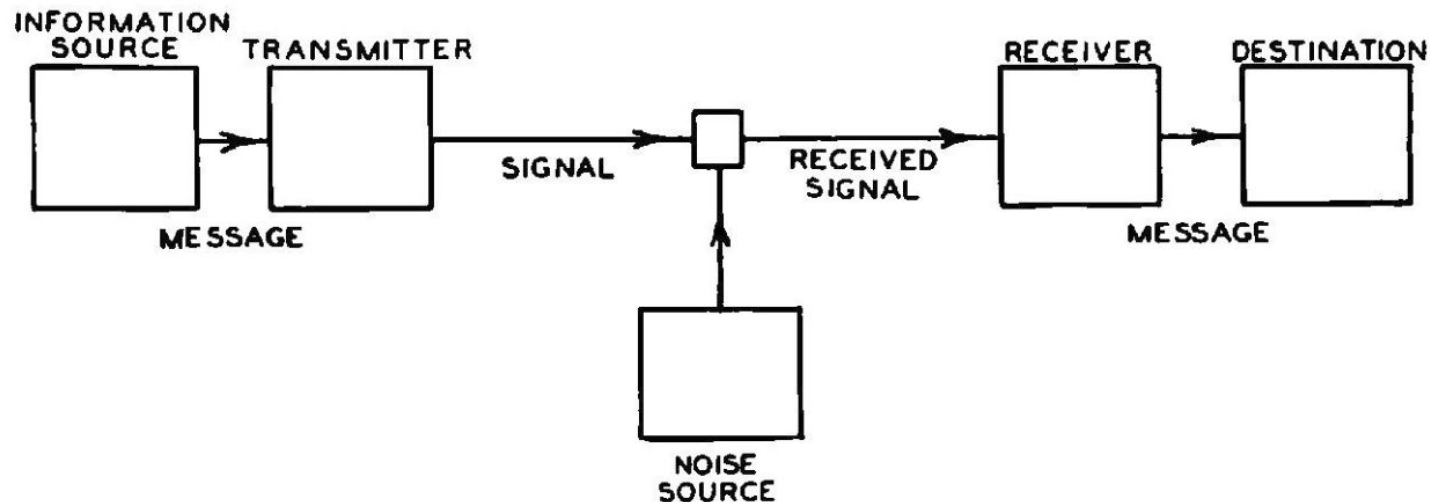
1993–2012

Claude Shannon



A Mathematical Theory of Communication (1948)

The noisy channel model



<https://www.quantamagazine.org/how-claude-shannons-information-theory-invented-the-future-20201222/>

The statistical revolution: 1990s

A Paradigm Shift in NLP

- **EE & ASR Influx:** Probabilistic modeling, corpus statistics, supervised learning, and empirical evaluation.
- **New Data Sources:** Explosion of machine-readable text and human-annotated data (e.g., the Penn Treebank).
- **Pragmatic Goals:** Focus on practical tasks like text categorization, POS tagging, NER, and parsing.
- **Core Tools:** Naïve Bayes classifiers, HMMs, PCFGs, and CRFs.

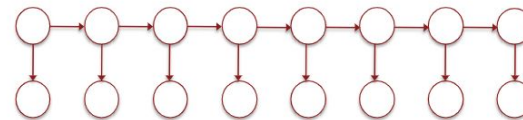
Fred Jelinek

Pioneer of statistical language processing and automatic speech recognition.



Fred Jelinek

Statistical Sequence Modeling (HMMs)



Probabilistic spelling correction

"She is a stellar and versatile actress whose combination of sass and glamour attracts"

Let's estimate probabilities of corrections following one another:

Option 1: "versatile actress whose"

$$P(\text{actress} \mid \text{versatile}) = .0019$$

$$P(\text{whose} \mid \text{actress}) = .0043$$

$$\begin{aligned} \text{Joint Probability: } P(\text{"versatile actress whose"}) \\ = .00019 \times .0043 = 817,000 \times 10^{-10} \end{aligned}$$

Option 2: "versatile across whose"

$$P(\text{across} \mid \text{versatile}) = .000092$$

$$P(\text{whose} \mid \text{across}) = .000026$$

$$\begin{aligned} \text{Joint Probability: } P(\text{"versatile across whose"}) \\ = .000092 \times .000026 = 239 \times 10^{-10} \end{aligned}$$

The rise of the machines: 2000s

Availability of much more powerful machines

- **Statistical Consolidation:** Integration of 1990s statistical gains with more sophisticated algorithms like MaxEnt, SVMs, Bayes Nets, and LDA.
- **Explosion of Big Data:** 100x web growth powered by the rapid deployment of massive server farms.
- **Paradigm Shift:** Learning focus transitioning from purely supervised to unsupervised methods.
- **Semantic Applications:** Revived and accelerated interest in higher-level semantic understanding.



Infrastructure Revolution

The 2000s marked the transition to industrial-scale computation, enabling the training of highly complex statistical models on unprecedented volumes of web data.

4. Deep Learning or Artificial Neural Networks for NLP

4a. Neural NLP

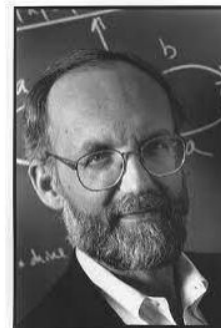
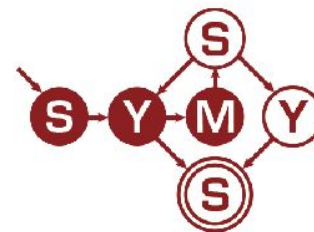
2013–2021

Symbolic AI vs. "Cybernetics"

Stanford, the home of "Symbolic Systems"

What are "Symbolic Systems"?

- **Study of Symbols:** Focuses on systems of meaningful symbols representing our world—such as human languages, logics, and programming languages.
- **The Processors:** Examines the physical and social systems operating on these symbols—like brains, computers, and complex social structures.
- **vs. Cognitive Science:** While cognitive science targets natural mind and intelligence, Symbolic Systems gives equal focus to human-constructed representational systems.



Jon Barwise
(1942–2000)

Symbolic systems versus their processors

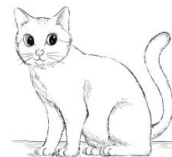
Symbolic Systems

- **Language as Archetype:** Language is the symbolic system par excellence; we should study and leverage its symbolic structure.
- **Processor Distinction:** This does not imply the human brain—the main symbol processor—is implemented as a physical symbol system.
- **Design Choice:** We need not design artificial NLP systems as physical symbol systems.

Connectionist Processors

- **Neural Brains:** The biological brain is structurally more akin to a neural network model than a symbolic computer.
- **Superior Scaling:** Artificial neural network models scale much better than symbolic counterparts.
- **Representational Capacity:** Continuous neural networks excel at capturing the rich, noisy world represented by symbolic systems.

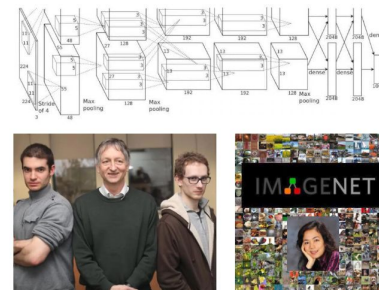
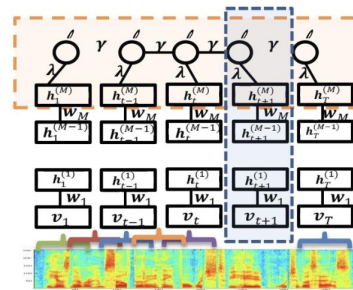
cat



Deep Learning Breakthroughs by Hinton... at the University of Toronto, 2009–2012

01. Speech Recognition (2009–2012)

Hinton's lab applied Deep Belief Networks and Deep Neural Networks to acoustic modeling, achieving massive word error rate reductions and transforming commercial speech systems.



02. ImageNet Revolution (2012)

AlexNet, a deep convolutional neural network trained on GPUs by Krizhevsky, Sutskever, and Hinton, won the ImageNet competition by a historic margin, launching the modern AI era.

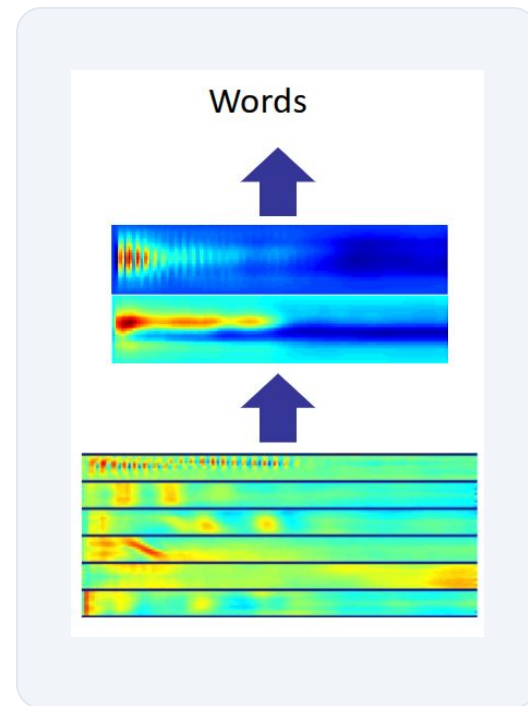
Key Milestones Illustrated:

- Acoustic DBM & DNN speech spectrograms (left)
- AlexNet CNN 8-layer architecture (top-right)
- The Toronto team: Sutskever, Hinton, Krizhevsky (center)
- ImageNet visual classification database (right)

Deep Learning for Speech Recognition

- **Probabilistic Success:** Speech recognition had first shown the breakthrough success of probabilistic methods: HMMs and GMMs.
- **Deep Learning Pivot:** The first breakthrough results of “deep learning” on large datasets also happened in speech recognition.
- **George Dahl et al. (2010/2012):** Context-Dependent Pre-trained Deep Neural Networks for Large Vocabulary Speech Recognition.

Acoustic model \ WER	RT03S FSH	Hub5 SWB
Traditional GMM (D. et al. 2012)	27.4	23.6
Deep Learning (Dahl et al. 2012)	18.5 (-33%)	16.1 (-32%)
Deep Learning (Saon et al. 2017)	8.0 (-71%)	5.5 (-77%)



“(Artificial) neural (network)” or “deep learning” models for word meaning

Vector Representations

Instead of treating words as atomic symbols, modern NLP models represent them as dense vectors of real numbers in a continuous vector space.

- **Vector of Numbers:** Each word is mapped to a list of coordinates capturing its syntactic and semantic properties.
- **Context-Driven:** These vectors are learned from large corpora based on the company each word keeps.
- **Semantic Proximity:** The core principle is that **similar vectors correspond to similar meanings**.

Visualizing Word Embeddings

$$\text{versatile} = \begin{pmatrix} 0.286 \\ 0.792 \\ -0.177 \\ -0.107 \\ 0.109 \\ -0.542 \\ 0.349 \\ 0.271 \end{pmatrix} \quad \text{Similar vectors} = \text{similar meaning}$$

A multi-dimensional vector representing the semantic profile of "versatile". In this continuous space, words with related meanings are geometrically close.

Learn vectors via distributional similarity

Foundational Quote

“You shall know a word by the company it keeps”

— J. R. Firth (1957: 11)

Core Principle

Distributional Similarity

Defining similarity via contextual distributions in texts is one of the most successful ideas of modern computational linguistics.

Linguistic Example

How do we learn word vectors?

We observe the contexts in which words appear. Consider this sample text:

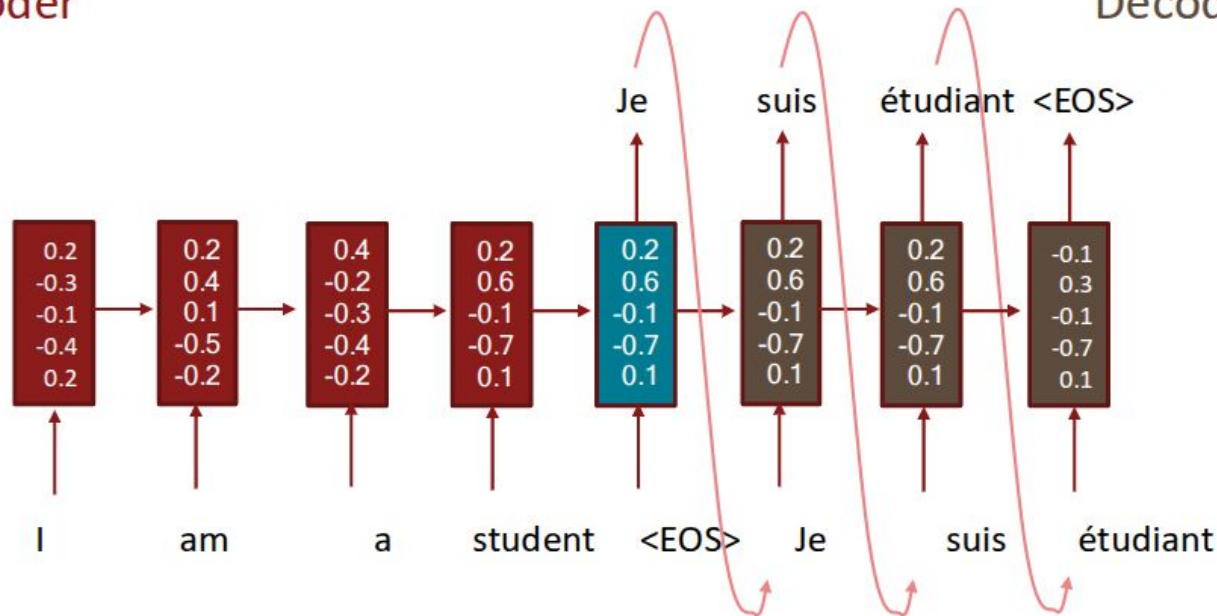
*“...any devices with a **web browser**, from **laptops** and **tablets** to **smart phones**. Users can download it for home **computers** or **laptops** from **Microsoft Update** website...”*

These highlighted co-occurring words construct the distributional profile that represents and defines the vector for **laptops**.

An RNN encoder-decoder network

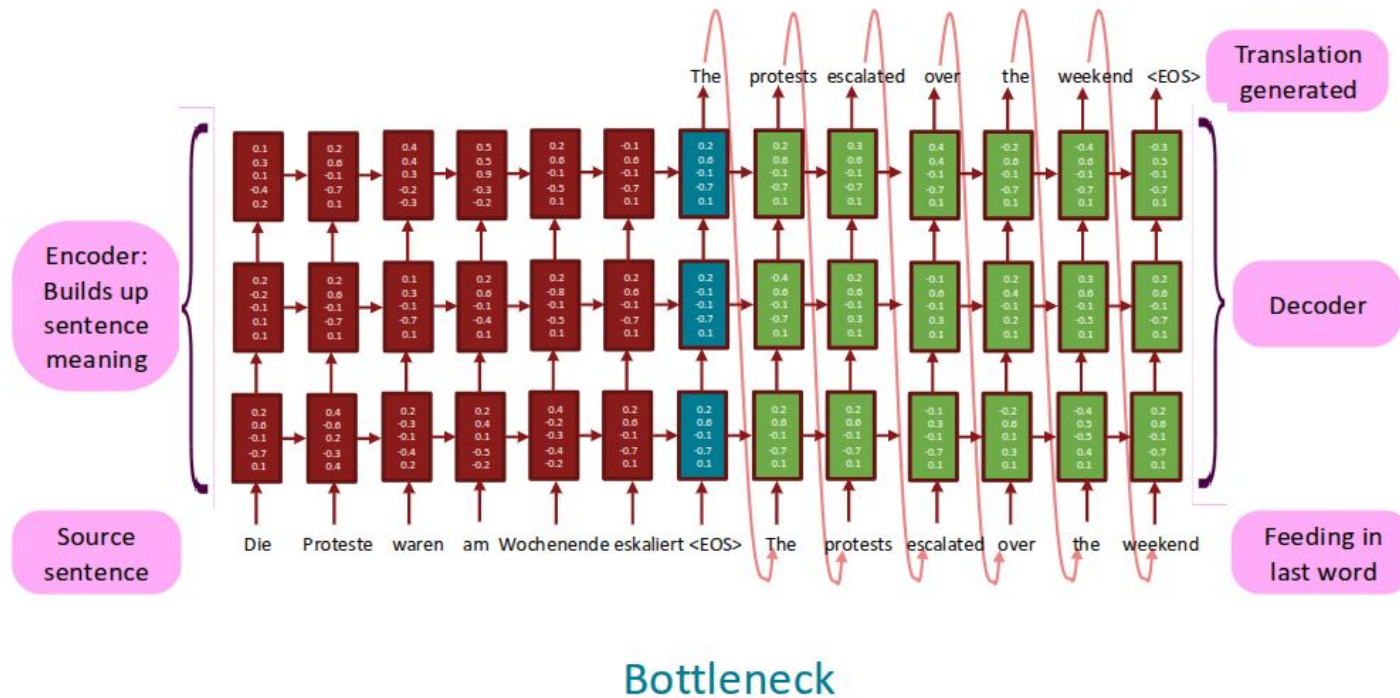
Encoder

Decoder



$$h_t = \tanh(W[x_t] + Uh_{t-1} + b)$$

An LSTM encoder-decoder network - [Sutskever et al. 2014]



MT Progress Over Time

[Edinburgh En-De WMT newstest2013 Cased BLEU; NMT 2015 from U. Montréal]

Key Milestones

2013 - 2014: SMT Era

Phrase-based and Syntax-based Statistical MT established the baseline BLEU scores below 20.3.

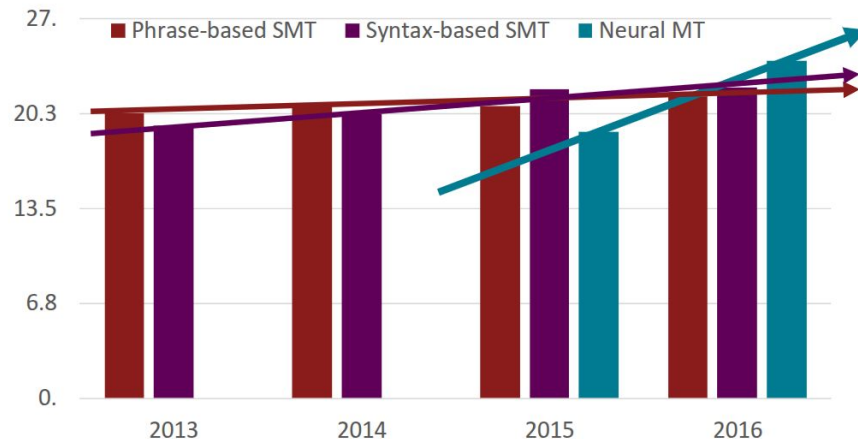
2015: Neural MT Arrival

Introduction of NMT (U. Montréal) initiated a steeper and more promising growth trajectory.

2016: The AI Breakthrough

NMT clearly outperformed traditional SMT, rapidly accelerating translation quality near 27 BLEU.

BLEU Score Growth Trend (En-De)



Source: http://www.meta-net.eu/events/meta-forum-2016/slides/09_sennrich.pdf

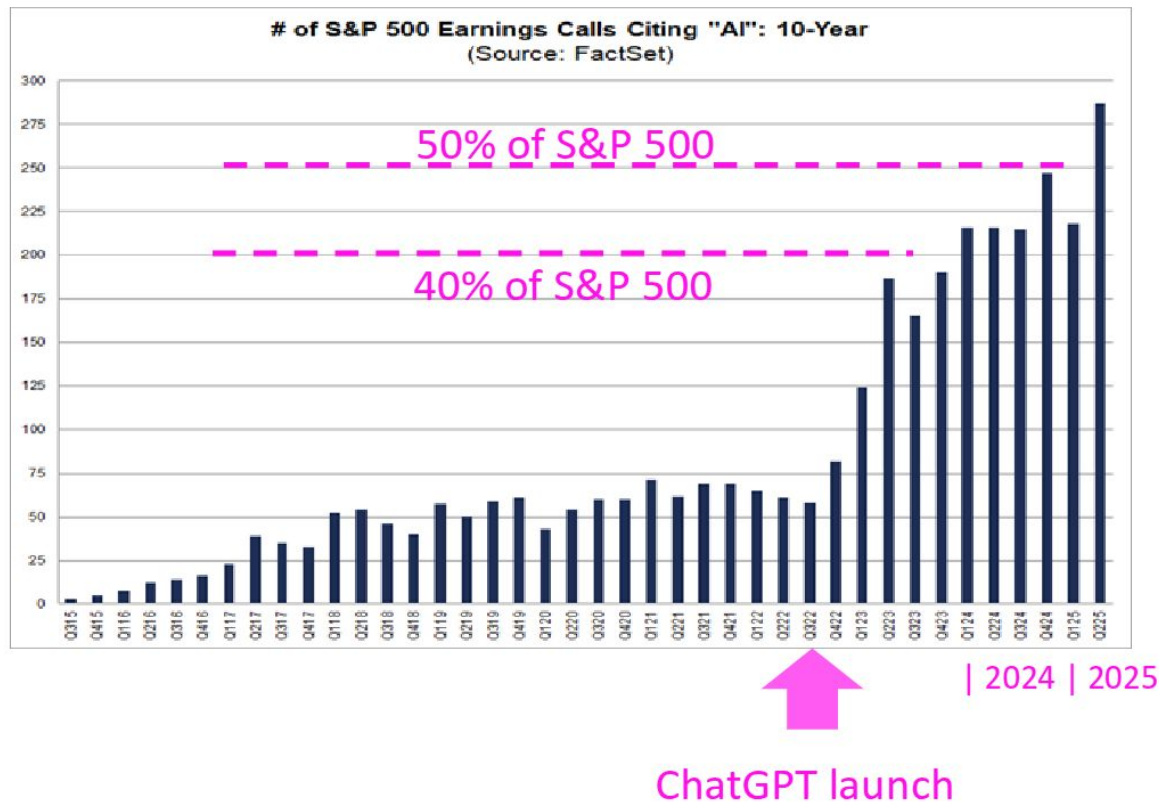
On Google Neural MT, see: <https://www.nytimes.com/2016/12/14/magazine/the-great-ai-awakening.html>

4. Deep Learning for NLP

4b. Large Language Models

2022 – Present

The Rise of Generative AI: S&P 500 Companies Discussing "AI"



The History of Language Models

1913**Andrey A. Markov**

Explores consonant-vowel transition probabilities in Alexander Pushkin's novel, Eugene Onegin, developing Markov Models.

1948**Claude E. Shannon**

Develops A Mathematical Theory of Information; explores character/word-level n-gram models, entropy, text generation.

1975**Frederick Jelinek's Group at IBM**

Defines and names the modern concept of a (probabilistic) "language model" for next token prediction. Used for spelling correction, speech recognition, machine translation...

LIMIT

But this technology wasn't considered a pathway to machines with artificial intelligence: For that we needed models of memory, knowledge representations, planning systems, and reasoning over abstract concepts.

The History of Language Models

1998

CPAT-Tree-Based Language Models (Chinese Text Verification)

ROCLing 1998 • First known use of "LLM" trigram • 200M word corpus

2000

A Neural Probabilistic Language Model (Bengio, Ducharme & Vincent)

NIPS 2000 • First neural language model built on 32 million token corpus, 31K vocab

2007

Large Language Models in Machine Translation (Brants et al.)

EMNLP 2007 • 2 trillion token corpus n-gram model of up to 5-grams

2018

GPT & BERT Era (Radford et al. / Devlin et al.)

3.3 billion token corpus • Transition to powerful Transformer-based representation models

2020–

Massive Parameter Scale (GPT-3, GPT-4, PaLM 2, Llama 3, Nemotron-4...)

100+ billion parameter models trained on > 1 trillion tokens

Advanced NLP - Session 1 - Introduction, History of NLP & Paradigm Shifts in LLMs - IUST

Course Overview

History of NLP